

Numerical Linear Algebra First-Year Exam, January 2016

Do 4 (four) problems.

1. Prove that every square matrix A has a Schur factorization.

2. Given a matrix $A \in \mathbb{C}^{m \times n}$, let

$$B = \begin{pmatrix} 0 & A^* \\ A & 0 \end{pmatrix} \quad \text{and} \quad C = A^*A.$$

- (a) Show that the singular values of A are the absolute values of eigenvalues of B .
- (b) Show that the singular values of A are the square roots of eigenvalues of C .
- (c) Assume now that A is square and invertible. Compute the two-norm condition numbers of B and C in terms of the two-norm condition number of A .
- (d) If A has condition number bigger than one, which has the larger condition number, B or C ?

3. This problem has two parts.

- (a) Compute $\det(\lambda I + uv^*)$ when $\lambda \in \mathbb{C}$, I is the $m \times m$ identity matrix, and $u, v \in \mathbb{C}^m$.
- (b) Prove necessary and sufficient conditions for $I + uv^*$ to be nonsingular and when it is, give a formula for its inverse.

4. This problem has two parts.

- (a) Given Cholesky decomposition of the Hermitian positive definite matrix $A = R^*R$, prove that $\|R\|_2 = \|R^*\|_2 = \|A\|_2^{1/2}$.
- (b) Now assume that B is a matrix that can be expressed as $B = T^*T$ for some upper triangular matrix T . Show that B is Hermitian and positive semi-definite, i.e. $x^*Bx \geq 0$ for all x .

5. Let $\{q_1, q_2, \dots, q_n\}$ be an orthonormal subset of $\mathbb{C}^m$. Show that

$$P = \sum_{i=1}^n q_i q_i^*$$

is an orthogonal projector with range equal to the span of $\{q_1, q_2, \dots, q_n\}$.